Introduction to Recurrent Neural Networks (RNNs)

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Outline of this Module

1. Introduction to Sequence Modelling
2. Introduction to Recurrent Neural Networks
3. Advanced Memory based Models
4. Modern Recurrent Neural Networks

The Sequential Data

Introduction to Sequence Modelling

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Key Concept

Analyzing this data involves understanding patterns, trends, and contexts **over time or space**.

Common Sequential data and Tasks

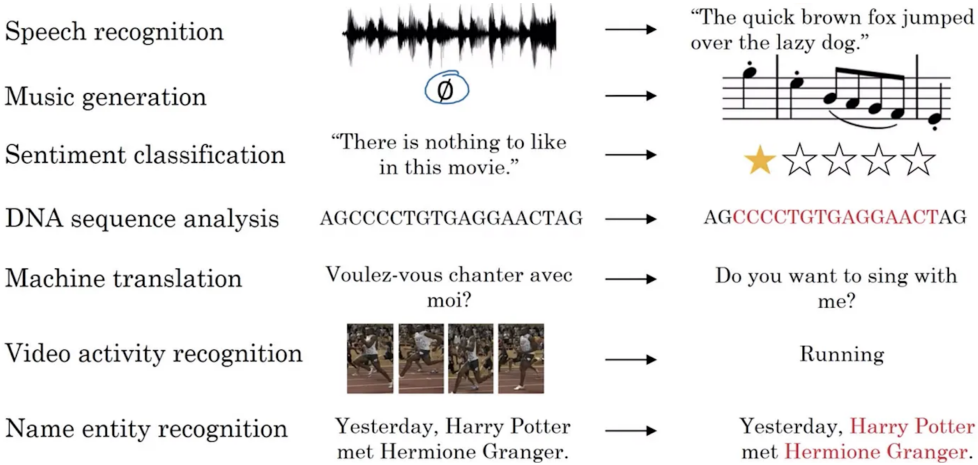

Figure: Common Sequential data and Tasks. (source: AIML.com)

Sequence Models

Mathematical Formulation

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Language Models

When a sequence model is applied to text data, it is called a **language model**. Its goal is to estimate the probability of a sequence of words or characters.

Autoregressive Models

Definition

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How it Works

To predict the next element x_t, the model uses the preceding sequence $(x_1, \ldots, x_{t-1})$ as input.

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First-Order Markov Model

- Assumes that the current state x_t only depends on the immediately preceding state x_{t-1}.
- The probability formula simplifies to:

$$P(x_1, \ldots, x_T) \approx P(x_1) \prod_{t=2}^{T} P(x_t \mid x_{t-1})$$

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- This can be extended to consider a fixed-length window of previous states (e.g., trigrams, etc.).

Linear Regression for Sequence modelling

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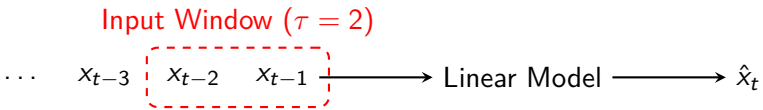

Diagram showing a linear model predicting x_t using a window of the two previous time steps.

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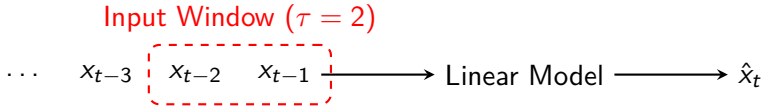

Diagram showing a linear model predicting x_t using a window of the two previous time steps.

Mathematical Formulation

The prediction, $\hat{x}_t$, is a weighted sum of the previous τ observations plus a bias term:

$$\hat{x}_t = \sum_{i=1}^{\tau} w_i x_{t-i} + b$$

- x_{t-i}: Past values (our features).
- w_i: Model weights to be learned.
- b: A bias term.

Limitations of Linear Regression for Sequence modelling

1. The Linearity Assumption

Linear models assume that the future is just a simple weighted sum of the past. They fail to capture complex, non-linear relationships, such as:

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- Sudden shocks or spikes in the data (e.g., a stock market crash).

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2. Lack of Long-Term Memory (Fixed Window)

By conditioning only on a fixed-length window (τ), the model is completely blind to any information that occurred before that window. This is a major drawback when long-range dependencies are important.

Limitations of Linear Regression for Sequence modelling

3. Error Accumulation

For k-step-ahead predictions (predicting $\hat{x}_{t+k}$), the model must use its own previous predictions as input. This creates a feedback loop where errors compound over time.

- The prediction $\hat{x}_{t+1}$ will have some error, ϵ_1.
- This incorrect value, $\hat{x}_{t+1}$, is then fed back into the model to predict $\hat{x}_{t+2}$.
- The error in the input leads to a new, often larger, error ϵ_2. This process repeats, causing the predictions to **diverge rapidly** from the true data.

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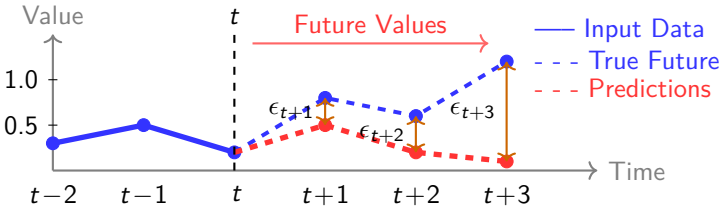

Limitations of MLPs and CNNs for Sequence Tasks

Why MLPs and CNNs are Not Ideal for Sequences

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We need a model that processes sequences step-by-step, with memory.

Language modelling

Text preprocessing tricks

From Text to Sequences

The Goal: Making Text Understandable to Machines

Neural networks don't understand words like "hello" or "world". They only understand numbers. Our first and most critical task is to convert our raw text corpus into a numerical format that a model can process.

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Raw Text	→	Tokens	→	Numerical Sequence
"The cat sat..."		["the", "cat", "sat", ...]		[1, 2, 3, ...]
Step 1		**Step 2**		**Step 3**
Read Data		Tokenize		Create Vocabulary

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The Core Idea

We will build a dictionary (a **vocabulary**) that maps every unique word or character to a unique integer index.

Step 1 & 2: Tokenization and Building a Vocabulary

Tokenization: Breaking Text into Pieces

The first step is to split the text into smaller units called **tokens**. We have two main choices:

- **Character Tokens:** Split the text into individual characters ('h', 'e', 'l', 'l', 'o'). This results in a small vocabulary but very long sequences.
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Building the Vocabulary

Find all unique tokens in the text corpus and then assign a unique integer index to each one.

- **Corpus:** "The cat sat on the wall."
- **Unique Word Tokens:** {"the", "cat", "sat", "on", "wall"}
- **Vocabulary (Vocab):** We create a mapping. It's also crucial to add a special token for any words that might appear later but are not in our current vocabulary.

Step 3: Creating the Final Numerical Sequence

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With our vocabulary complete, we can now convert any sentence into a sequence of integers. We simply replace each token with its corresponding index from the vocabulary.

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Example Conversion

- **Vocabulary:** `{"<unk>": 0, "the": 1, "cat": 2, "sat": 3, "on": 4, "wall": 5}`

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- **Sentence 1:** "the cat sat on the wall"
- **Numerical Sequence:** `[1, 2, 3, 4, 1, 5]`
- **Sentence 2:** "the cat sat on the chair"
- *Note: "chair" is not in our vocabulary!*
- **Numerical Sequence:** `[1, 2, 3, 4, 1, 0]` (*Here, "chair" is mapped to the <unk> token index, 0*)

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The Chain Rule of Probability

We can factor this joint probability into a product of conditional probabilities. This is the core of modern autoregressive language models.

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The Challenge

The conditioning term $P(x_t \mid x_1, \ldots, x_{t-1})$ becomes computationally expensive and statistically difficult to model as the sequence length t grows.

Partitioning Sequences

Preparing Data for Neural Models

To train a neural language model, we need to create feature-label pairs from our long text sequence. There are two common strategies – Random Sampling and Sequential Partitioning.

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Random Sampling

- To introduce randomness, a random number of tokens is discarded from the start of the corpus at the beginning of each epoch.
- The rest of the corpus is then partitioned into multiple non-overlapping subsequences of a fixed length, τ.
- For any given input subsequence $X = (x_1, x_2, \ldots, x_\tau)$, the target (label) sequence Y is the same sequence shifted by one token to the left: $Y = (x_2, x_3, \ldots, x_{\tau+1})$.
- Each subsequence created this way is treated as an independent example for training.

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Input-Target Pairs:

Input1: sat | on | the

Target1: on | the | mat

$X = (x_1, x_2, x_3)$
$Y = (x_2, x_3, x_4)$

Input2: mat | and | slept

Target2: and | slept | all

$X = (x_4, x_5, x_6)$
$Y = (x_5, x_6, x_7)$

Each subsequence = Independent training example

Perplexity

How Good Is a Language Model?

Perplexity (PPL) is the standard metric for evaluating language models. It measures how "surprised" or "perplexed" a model is by a sequence of text.

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Mathematical Definition

Perplexity is the exponentiation of the average negative log-likelihood of the sequence. For a sequence of length T:

$$\text{PPL} = \exp\left(-\frac{1}{T}\sum_{t=1}^{T}\log P(x_t \mid x_1, \ldots, x_{t-1})\right)$$

Perplexity

Intuition

A perplexity of k means that, on average, the model is as uncertain about the next word as if it had to choose uniformly from k different words.

- If PPL $= 100$, the model is as confused as someone randomly guessing from 100 words.
- If PPL $= 10$, the model has narrowed down the possibilities to about 10 words.

Recurrent Neural Networks

Memory based models for sequential data

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The RNN Solution: A Latent Hidden State

Instead of conditioning on the raw sequence of past tokens, an RNN maintains a compact summary of the entire history in a **hidden state**, h_t.

- The model approximates the true probability with the hidden state:

$$P(x_t \mid x_1, \ldots, x_{t-1}) \approx P(x_t \mid h_{t-1})$$

- This state is updated at each time step using the current input x_t and the previous state h_{t-1}:

$$h_t = f(x_t, h_{t-1})$$

The Recurrent Neural Network

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  - Updates its hidden state h_t
  - Produces an output y_t

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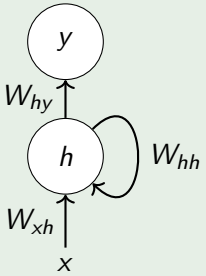

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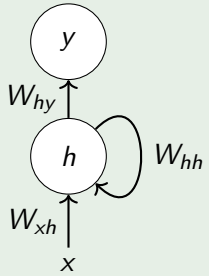
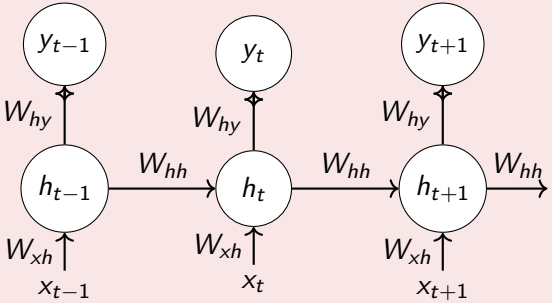
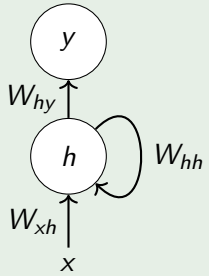


Unfolding Over Time



A Basic Recurrent neuron

The Recurrent neuron



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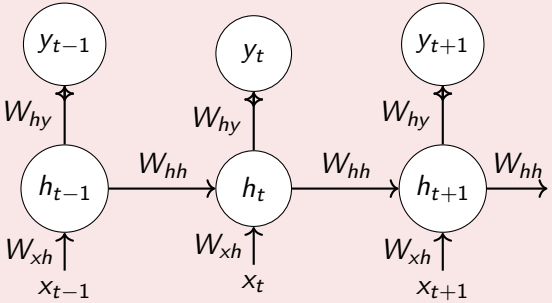

Key Point: The same weight matrix is used at every time step.

A Layer of Recurrent Neurons

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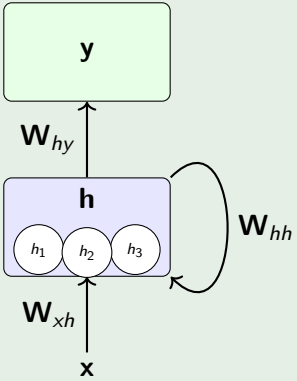
RNN Layer (Folded)



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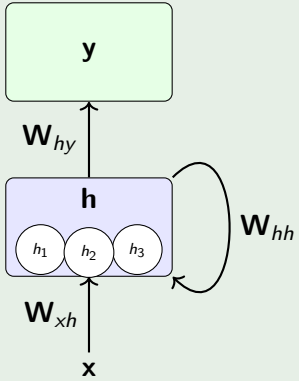
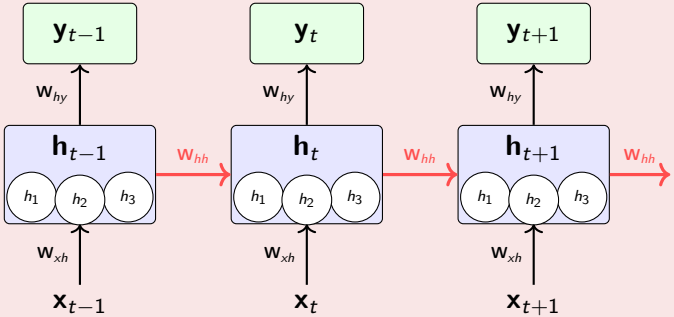


Unfolded RNN Layer Over Time



RNN Computation in a neuron

The Weights of a Single Recurrent Neuron

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For one neuron, the computation is:

$$h_{(t)} = \tanh(w_{xh}x_{(t)} + w_{hh}h_{(t-1)} + b_h)$$

and

$$y_{(t)} = w_{hy}h_{(t)} + b_y$$

RNN Computations in a Layer

From Weights to Weight Matrices: The RNN Layer

An RNN layer consists of many neurons. To compute efficiently, we group the individual weights from all neurons into shared weight matrices.

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This allows us to process an entire vector of inputs $\mathrm{x}_{(t)}$ and update the entire layer's hidden state vector $\mathrm{h}_{(t)}$ at once with the following equations:

$$\mathrm{h}_{(t)} = \tanh(\mathbf{W}_{xh}\mathrm{x}_{(t)} + \mathbf{W}_{hh}\mathrm{h}_{(t-1)} + \mathrm{b}_h)$$
$$\mathrm{y}_{(t)} = \mathbf{W}_{hy}\mathrm{h}_{(t)} + \mathrm{b}_y$$

Numerical Example: Next-Character Prediction

The Task

Given the input character "e" and the memory of having just seen "h", predict the next character. Our simple vocabulary is `{'h':0, 'e':1, 'l':2, 'o':3}`.

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- Input $x_{(t)}$ (char 'e'): $\begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix}^T$
- Groundtruth Target $y_{(t)}$ (char 'l'): $\begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix}^T$

Numerical Example: Next-Character Prediction

Setup

- Previous State $\mathrm{h}_{(t-1)}$ (from char 'h'): $\begin{pmatrix} 0.6 \\ 0.2 \end{pmatrix}$ *(Pre-computed for simplicity)*
- Weights & Biases: $\mathbf{W}_{xh} = \begin{pmatrix} 0.1 & 0.4 \\ 0.2 & 0.5 \\ 0.3 & 0.6 \\ 0.4 & 0.7 \end{pmatrix}$, $\mathbf{W}_{hh} = \begin{pmatrix} 0.1 & 0.5 \\ 0.5 & 0.1 \end{pmatrix}$, $\mathbf{W}_{hy} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & 3 \\ 0 & -2 \end{pmatrix}$,

  $\mathrm{b}_h = \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix}$, $\mathrm{b}_y = \begin{pmatrix} 0.1 \\ 0.2 \\ 0.3 \\ 0.4 \end{pmatrix}$

Numerical Example: Next-Character Prediction

Step 1: Calculate the New Hidden State $\mathbf{h}_{(t)}$

$$\mathrm{h}_{(t)} = \tanh(\mathbf{W}_{xh}^{T}\mathrm{x}_{(t)} + \mathbf{W}_{hh}\mathrm{h}_{(t-1)} + \mathrm{b}_{h})$$

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The output $\mathrm{y}_{(t)}$ gives a score (logit) for each character in the vocabulary.

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The highest logit is 2.89, which corresponds to the character 'l'. **The model predicts 'l'!**

Backpropagation in RNNs: The Challenge

The Problem: Dependencies Through Time

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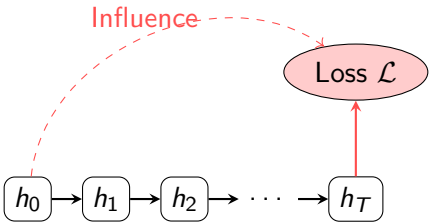
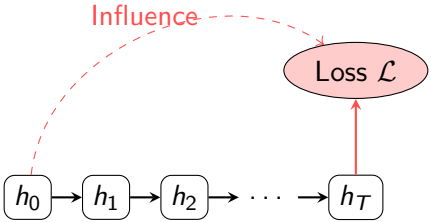


The Main Challenge



The Core Issue

To calculate the gradient of the loss with respect to early parameters (like the weights used to compute h_1), we must backpropagate "through time", applying the chain rule across the entire sequence.

Backpropagation Through Time (BPTT)

What is BPTT?

BPTT is simply the application of the standard backpropagation algorithm to the "unrolled" computational graph of an RNN. It is not a new algorithm, but a name for the process.

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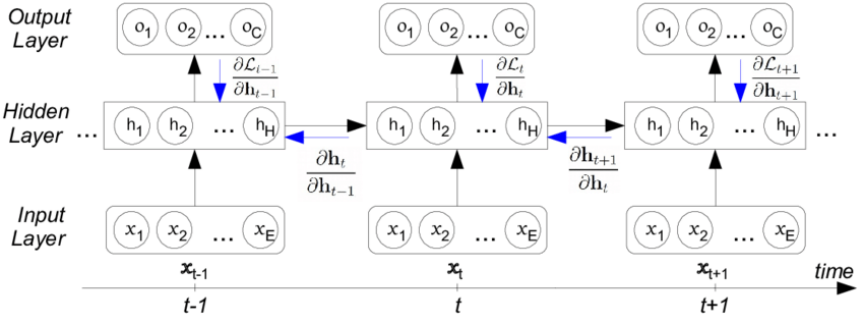

Figure: BPTT Illustration (source: DOI - 10.13140/RG.2.2.34411.08483)

Backpropagation Through Time (BPTT)

The Chain Rule in BPTT

The gradient of the final loss $\mathcal{L}$ with respect to the hidden state at time t, $\frac{\partial \mathcal{L}}{\partial h_t}$, has two components:

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$$\frac{\partial \mathcal{L}}{\partial h_t} = \underbrace{\frac{\partial \mathcal{L}}{\partial y_t}\frac{\partial y_t}{\partial h_t}}_{\text{Current Loss}} + \underbrace{\frac{\partial \mathcal{L}}{\partial h_{t+1}}\frac{\partial h_{t+1}}{\partial h_t}}_{\text{Future Loss}}$$

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This recursive formula is the heart of BPTT.

The Mathematics of Backpropagation Through Time

Objective

Our goal is to compute the gradients of the total loss $\mathcal{L} = \sum_{t=1}^{T} \mathcal{L}_t$ with respect to the model parameters: $\mathbf{W}_{xh}, \mathbf{W}_{hh}, \mathbf{W}_{hy}, \mathbf{b}_h, \mathbf{b}_y$. We use the chain rule to backpropagate the error from the end of the sequence to the beginning.

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Step 1: The Chain Rule for Gradients

First, we compute the gradient of the loss with respect to the output and the hidden state at each time step t.

$$\frac{\partial \mathcal{L}}{\partial \mathbf{y}_t} = \frac{\partial \mathcal{L}_t}{\partial \mathbf{y}_t} \qquad \text{(From the loss function)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} = \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{y}_t}\frac{\partial \mathbf{y}_t}{\partial \mathbf{h}_t}}_{\text{From current output}} + \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_{t+1}}\frac{\partial \mathbf{h}_{t+1}}{\partial \mathbf{h}_t}}_{\text{From next hidden state}}$$

The Mathematics of Backpropagation Through Time

Step 2: Calculating Parameter Gradients

Once we have $\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t}$, we can find the gradients for the weights and biases at that time step. The final gradient for a parameter is the sum of its gradients from all time steps.

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_{hy}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}}{\partial \mathbf{y}_t} \frac{\partial \mathbf{y}_t}{\partial \mathbf{W}_{hy}} = \sum_{t=1}^{T} \left(\frac{\partial \mathcal{L}_t}{\partial \mathbf{y}_t} \right)^T \mathbf{h}_t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_{xh}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} \frac{\partial \mathbf{h}_t}{\partial \mathbf{W}_{xh}} = \sum_{t=1}^{T} \text{diag}(\phi'(\dots))^T \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} \right) \mathbf{x}_t^T$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_{hh}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} \frac{\partial \mathbf{h}_t}{\partial \mathbf{W}_{hh}} = \sum_{t=1}^{T} \text{diag}(\phi'(\dots))^T \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} \right) \mathbf{h}_{t-1}^T$$

BPTT: Numerical Example

Setup

Consider a 1-neuron RNN with a sequence of length 2. Let loss $\mathcal{L}_t = (y_t - \text{target}_t)^2$.

- Inputs: $x_1 = 2, x_2 = 3$
- Initial State: $h_0 = 0$
- Weights: $W_{xh} = 0.5, W_{hh} = 0.2, W_{hy} = 0.4$ (Biases are 0 for simplicity)
- Activation: $\phi = \tanh$, so $\phi'(z) = 1 - \tanh^2(z)$
- Targets: $\text{target}_1 = 0.5, \text{target}_2 = 0.8$

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Step 1: Forward Pass

- **t=1**: $h_1 = \tanh(0.5 \cdot 2 + 0.2 \cdot 0) = \tanh(1) = 0.76$. $y_1 = 0.4 \cdot 0.76 = 0.304$.
  $\mathcal{L}_1 = (0.304 - 0.5)^2 = 0.038$.
- **t=2**: $h_2 = \tanh(0.5 \cdot 3 + 0.2 \cdot 0.76) = \tanh(1.652) = 0.93$. $y_2 = 0.4 \cdot 0.93 = 0.744$.
  $\mathcal{L}_2 = (0.744 - 0.8)^2 = 0.003$.

BPTT: Numerical Example

Step 2: Backward Pass (starting from the end)

Total Loss $\mathcal{L} = 0.038 + 0.003 = 0.041$.

- **t=2**: $\frac{\partial \mathcal{L}}{\partial y_2} = 2(y_2 - \text{target}_2) = 2(0.744 - 0.8) = -0.112$.
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BPTT: Numerical Example

Step 2: Backward Pass (starting from the end)

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- **t=1**: $\frac{\partial \mathcal{L}}{\partial y_1} = 2(y_1 - \text{target}_1) = 2(0.304 - 0.5) = -0.392$.
  The crucial step:
  $\frac{\partial \mathcal{L}}{\partial h_1} = (\frac{\partial \mathcal{L}}{\partial y_1} W_{hy}) + (\frac{\partial \mathcal{L}}{\partial h_2}\frac{\partial h_2}{\partial h_1})$. $\frac{\partial h_2}{\partial h_1} = (1 - h_2^2)W_{hh} = (1 - 0.93^2) \cdot 0.2 = 0.027$.
  $\frac{\partial \mathcal{L}}{\partial h_1} = (-0.392 \cdot 0.4) + (-0.045 \cdot 0.027) = -0.1568 - 0.0012 = -0.158$.
  $\frac{\partial \mathcal{L}_1}{\partial W_{hh}} = \frac{\partial \mathcal{L}}{\partial h_1}(1 - h_1^2)h_0 = -0.158 \cdot (1 - 0.76^2) \cdot 0 = 0$.

Final Gradient: $\frac{\partial \mathcal{L}}{\partial W_{hh}} = \frac{\partial \mathcal{L}_1}{\partial W_{hh}} + \frac{\partial \mathcal{L}_2}{\partial W_{hh}} = 0 - 0.0046 = -0.0046$.

Numerical Instability in BPTT

The Problem with Long Sequences

For a sequence of length T, calculating the gradient for the initial weights involves a chain rule product of T matrices.

$$\frac{\partial h_T}{\partial h_0} = \prod_{t=1}^{T} \frac{\partial h_t}{\partial h_{t-1}} = \prod_{t=1}^{T} \mathbf{W}_{hh}^T \text{diag}(\phi'(\ldots))$$

This long product is the source of major numerical instability.

The Twin Problems: Exploding and Vanishing Gradients

Gradient Explosion

- **What happens?** If the recurrent weight matrix $\mathbf{W}_{hh}$ has large values, the long product of matrices can grow exponentially.

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Truncated BPTT: A Practical Compromise

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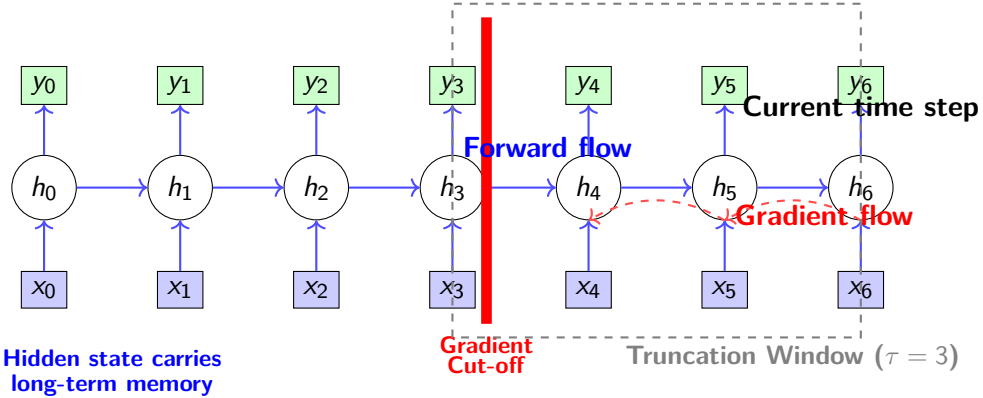
The Solution: Truncated BPTT

Instead of backpropagating through the entire sequence, we detach the gradient history after a fixed number of steps (τ).

- The model still propagates its hidden state forward through the whole sequence, maintaining its long-term memory.
- However, during the backward pass, the gradient calculation is cut off after τ steps.

Truncated BPTT: Illustration



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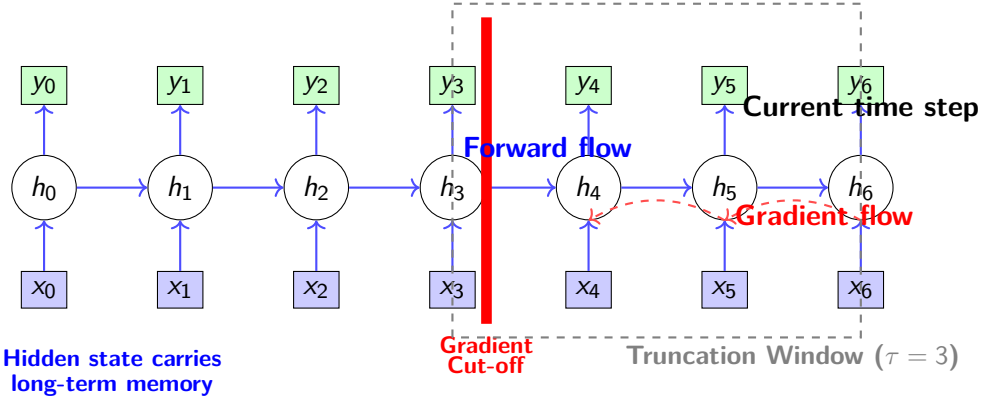

Trade-off

It limits the model's ability to learn dependencies that span longer than τ time steps.

Advanced Memory based Models

Introduction to LSTM and GRU

Training RNNs: Challenges

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- Hence it results in Training unstability (often resulting in NaN values).

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LSTM Architecture: The Gates

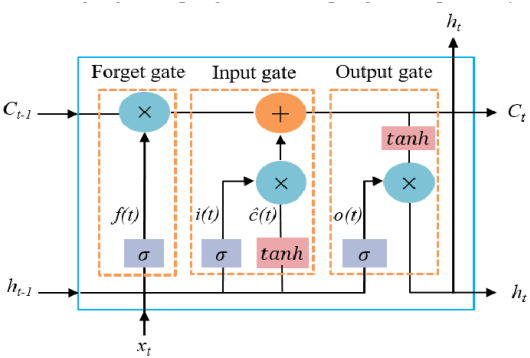

Figure: LSTM architecture (source: DOI-10.1109/ACCESS.2021.3125733)

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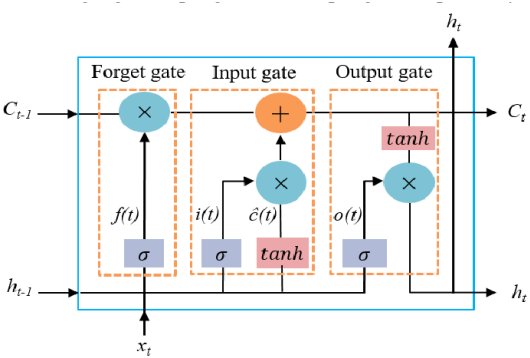

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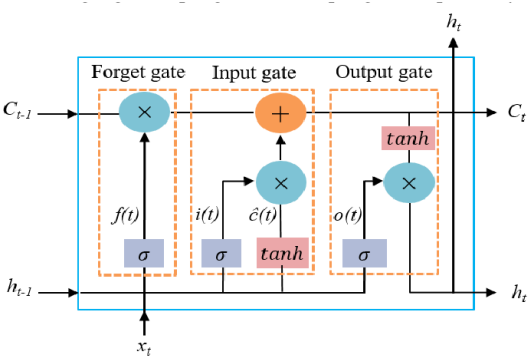

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Function and Intuition

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- **Interpretation:** The forget gate learns to selectively discard irrelevant past information based on the current input and previous hidden state.

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  - These two are multiplied element-wise to form the new information that will be added to the cell state.

LSTM Equations: Updating the Cell State

Function and Intuition

This is the core of the LSTM's memory mechanism. The new cell state $\mathbf{c_t}$ is a combination of the old cell state, which has been partially forgotten, and the new information, which has been carefully selected.

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- **Interpretation:** This additive interaction is crucial for preserving gradients. Instead of a series of multiplicative steps, the addition allows gradients to flow directly back, preventing them from vanishing.

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  - This tanh output is then multiplied element-wise by the output gate $\mathbf{o_t}$, which selectively exposes the useful information from the cell state.

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Benefits

Due to its simpler design, a GRU often trains faster than an LSTM and can perform just as well, or sometimes even better, on less complex datasets.

GRU architecture: Gates

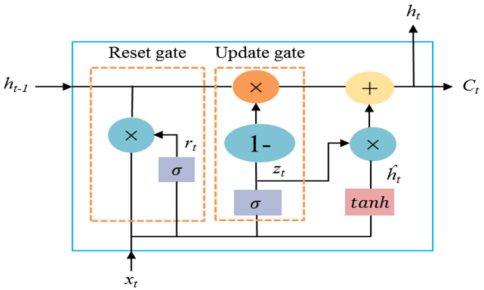

Figure: GRU Architecture (source: DOI-10.1007/s11042-023-15571-y)

The Core Components: Gating Mechanisms

- **Reset Gate (R_t):** Controls how much of the past information to forget.
- **Update Gate (Z_t):** Controls how much of the past information to carry forward.

GRU Architecture: The Reset Gate

Function: Controlling Short-Term Memory

The **Reset Gate ($\mathbf{R}_t$)** determines how the previous hidden state $\mathbf{H}_{t-1}$ is used to compute the *new candidate hidden state*.

- When entries in $\mathbf{R}_t$ are close to 1, the gate allows the previous hidden state to pass through, preserving its memory.
- When entries are close to 0, the gate effectively "resets" or ignores parts of the previous hidden state. This is useful when the current input is the start of a new, unrelated sequence of information.

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Formula

$$\mathbf{R}_t = \sigma(\mathbf{X}_t\mathbf{W}_{xr} + \mathbf{H}_{t-1}\mathbf{W}_{hr} + \mathbf{b}_r)$$

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The **Update Gate ($\mathbf{Z}_t$)** controls how much of the previous hidden state $\mathbf{H}_{t-1}$ is directly carried over to the final hidden state $\mathbf{H}_t$.

- It acts similarly to the forget gate in an LSTM.
- When entries in $\mathbf{Z}_t$ are close to 1, the corresponding dimension in $\mathbf{H}_{t-1}$ is almost entirely copied to $\mathbf{H}_t$, preserving long-term memory.
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$$\mathbf{Z}_t = \sigma(\mathbf{X}_t\mathbf{W}_{xz} + \mathbf{H}_{t-1}\mathbf{W}_{hz} + \mathbf{b}_z)$$

GRU Architecture: Hidden State

Step 1: The Candidate Hidden State ($\tilde{\mathbf{H}}_t$)

First, a candidate hidden state is created. It's similar to the hidden state in a simple RNN, but with a key difference: the reset gate $\mathbf{R}_t$ controls how much of the previous state $\mathbf{H}_{t-1}$ influences this new candidate.

$$\tilde{\mathbf{H}}_t = \tanh(\mathbf{X}_t \mathbf{W}_{xh} + (\mathbf{R}_t \odot \mathbf{H}_{t-1})\mathbf{W}_{hh} + \mathbf{b}_h)$$

The $\odot$ symbol represents the element-wise (Hadamard) product.

GRU Architecture: Hidden State

Step 2: The Final Hidden State ($\mathbf{H}_t$)

The final hidden state is a linear interpolation between the previous hidden state $\mathbf{H}_{t-1}$ and the new candidate hidden state $\tilde{\mathbf{H}}_t$. The update gate $\mathbf{Z}_t$ determines the mixing ratio.

$$\mathbf{H}_t = \mathbf{Z}_t \odot \mathbf{H}_{t-1} + (1 - \mathbf{Z}_t) \odot \tilde{\mathbf{H}}_t$$

This mechanism allows the GRU to either retain old information (if $\mathbf{Z}_t \approx 1$) or fully replace it with new information (if $\mathbf{Z}_t \approx 0$).

Modern Recurrent Neural Networks

Introduction to Bidirectional models and Encoder-Decoder architecture

Introduction to Bidirectional RNNs

The Limitation of Standard RNNs

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- This is a major drawback for tasks where future context is crucial.
- **Example:** In the sentence "`The man who ___ movies is a critic`", we need the word "movies" to predict "reviews". A forward-only RNN would struggle to guess this from "The man who".

Introduction to Bidirectional RNN

The Solution: Look Both Ways

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- For any time step t, the final hidden state is formed by concatenating the hidden states from both the forward and backward layers.

Training of Bidirectional RNNs

Training a Bi-RNN

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is computed and loss is backpropagated

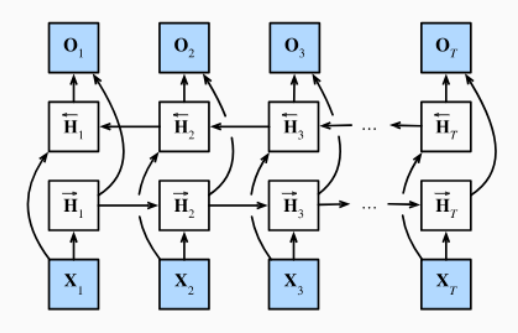


Figure: Bidirectional RNN. (source: D2L.ai)

Bidirectional RNNs: Illustrative Walkthrough

The Task: Predicting a Missing Word

Our goal is to understand how a Bidirectional RNN predicts the missing word in the sentence below. To make an accurate guess, the model needs to understand the context from both before and after the blank space.

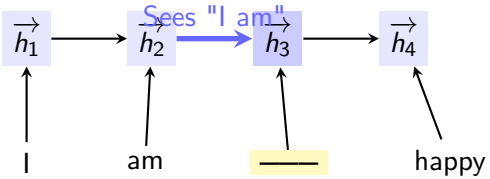
I am ——— happy

Let's walk through the process step-by-step.

Step 1: The Forward Pass (Past Context)

Reading from Left to Right

First, a standard RNN processes the sequence from the beginning. When it reaches the blank space at time step 3, its hidden state ($\overrightarrow{h_3}$) has only processed the past context: "I am".



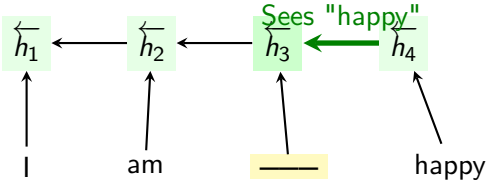
Limitation

This information alone is not enough to make a good prediction.

Step 2: The Backward Pass (Future Context)

Reading from Right to Left

Next, a second, independent RNN processes the sequence in reverse. Its hidden state at the blank space ($\overleftarrow{h}_3$) has only processed the future context: "happy".



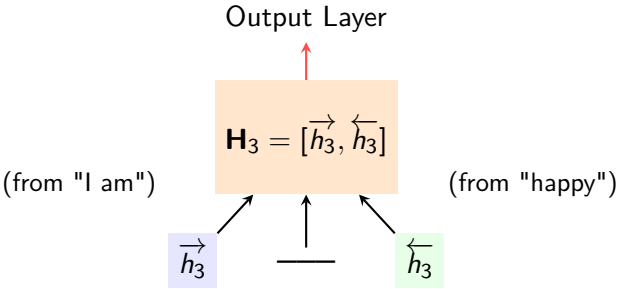
Limitation

This future context is helpful, but still incomplete on its own.

Step 3: Combining Context for Prediction

The Final Step: Concatenation

The Bidirectional RNN concatenates the hidden states from both the forward and backward passes. This combined state, $\mathbf{H}_3 = [\overrightarrow{h_3}, \overleftarrow{h_3}]$, now contains a rich representation of the full sentence context.



Encoder-Decoder Architectures

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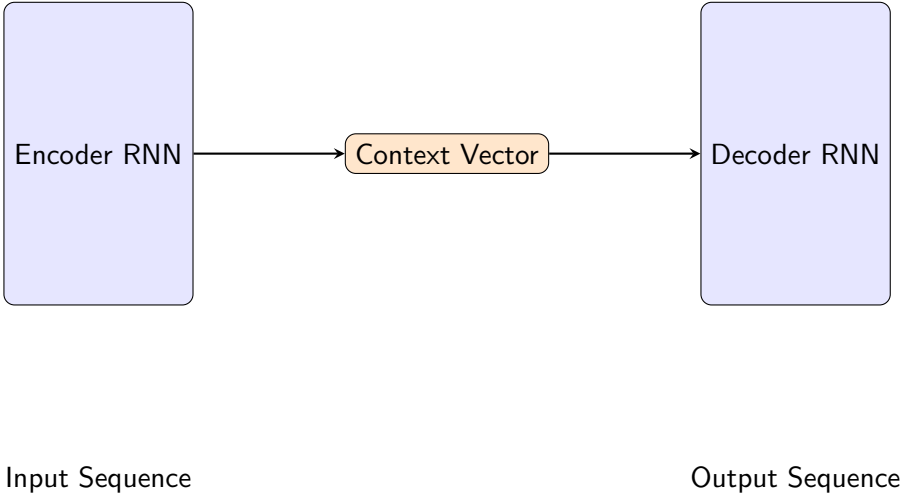
The Encoder-Decoder Framework

This architecture, also known as a Sequence-to-Sequence (Seq2Seq) model, is designed specifically for this problem. It consists of two main RNN components:

- **The Encoder:** An RNN that reads the entire input sequence step-by-step. Its sole purpose is to compress the information of the whole sequence into a single fixed-size vector, often called the **context vector**.
- **The Decoder:** Another RNN that is initialized with the encoder's context vector. It generates the output sequence one token at a time, using the context and the previously generated token.

Encoder-Decoder Architectures



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- At inference time, however, the model must use its own previously generated tokens as input, since the ground truth is not available.
- The use of Context Vector in a decoder follows a design choice, but in most cases it is concatenated with the input sequences to the decoder model.

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- **Masking:** When working with batches of sequences that have different lengths, padding is used to make them uniform. The loss function uses a mask to ignore these padding tokens during calculation, ensuring they do not contribute to the gradient updates.

Evaluating Predicted Sequences: The BLEU Score

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- If the predicted sequence is identical to the target, the BLEU score is 1.

BLEU Score: A Numerical Example

Let's consider an example with $k = 2$ (we evaluate unigrams and bigrams).

- **Target (Label) Sequence**: "A B C D E F"
- **Predicted Sequence**: "A B"

1. Calculate n-gram Precision (p_n):

- **Unigram Precision (p_1):**
  - Predicted unigrams: {'A', 'B'}
  - All are found in the target sequence.
  - $p_1 = \frac{\text{matched}}{\text{total}} = \frac{2}{2} = 1.0$
- **Bigram Precision (p_2):**
  - Predicted bigrams: {'A B'}
  - The bigram is found in the target sequence.
  - $p_2 = \frac{\text{matched}}{\text{total}} = \frac{1}{1} = 1.0$

Without a penalty, the score would be $(1.0 \times 1.0)^{1/2} = 1.0$, which is misleadingly perfect.

Applying the penalty term in BLEU

2. Apply the Brevity Penalty (BP):

- The predicted sequence is shorter than the target ($\text{len}_{\text{pred}} = 2$, $\text{len}_{\text{label}} = 6$).
- We must apply the penalty:

$$BP = \exp\left(1 - \frac{\text{len}_{\text{label}}}{\text{len}_{\text{pred}}}\right) = \exp\left(1 - \frac{6}{2}\right) = \exp(-2) \approx 0.135$$

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3. Final BLEU Score:

$$\text{BLEU} = BP \cdot (p_1 \cdot p_2)^{1/2} \approx 0.135 \times (1.0 \times 1.0)^{1/2} = 0.135$$

Conclusion: The brevity penalty correctly lowers the score from a perfect 1.0 to 0.135, reflecting that while the precision is high, the prediction is far too short.

Applications of Encoder-Decoder Architectures

Where Seq2Seq Models Shine

The Encoder-Decoder framework is the foundation for many state-of-the-art NLP and sequence-based tasks where the input and output lengths differ.

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- **Text Summarization** (e.g., reading a 500-word article and generating a 50-word summary).
- **Question Answering** (e.g., reading a context paragraph and generating a concise answer).

The Challenge of Sequence Prediction

From Greedy to Optimal Search

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  - **Con**: The cost is very high, which is computationally infeasible for any practical application.

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4. This process is repeated until an end-of-sequence token is generated or a maximum length is reached.

Beam Search: An Illustration

Beam Size, k=2

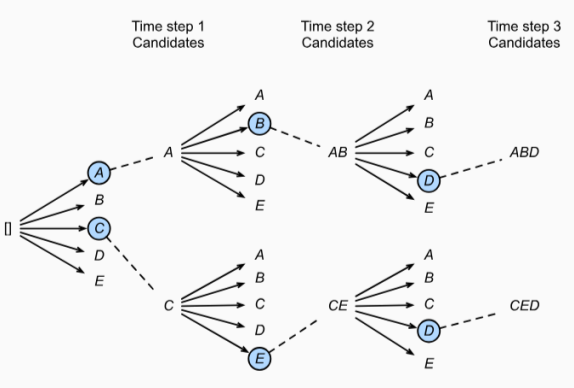

Figure: In the above example candidate solutions are: {A}, {C}, {A,B}, {C,E}, {A,B,D}, {C,E,D}(source: d2l.ai)

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where:

- L is the length of the candidate sequence.
- α is a hyperparameter (e.g., 0.75) that controls the strength of the length penalty. The term L^α in the denominator penalizes long sequences.

Final Selection and Scoring

- After the search is complete, we have a set of final candidate sequences.
- We choose the candidate sequence that maximizes the joint probability distribution.
- A simple product of probabilities is a way, but it is biased towards shorter sequences.
- To counteract this, the final score is often normalized by the sequence length:

$$\text{Score} = \frac{1}{L^\alpha} \log P(y_1, \ldots, y_L | c) = \frac{1}{L^\alpha} \sum_{t'=1}^{L} \log P(y_{t'} | y_1, \ldots, y_{t'-1}, c)$$

where:

- L is the length of the candidate sequence.
- α is a hyperparameter (e.g., 0.75) that controls the strength of the length penalty. The term L^α in the denominator penalizes long sequences.

Note

Greedy search is a special case of beam search arising when the beam size is set to 1.